bias_trainer.py

`src/moe_pruning/training/bias_trainer.py · 352 lines · module moe_pruning.training.bias_trainer`

What it does

Trains only the bias vectors so that each pruned layer reproduces the original layer's output on calibration inputs (layer-level mean squared error).

How it works

- `capture_layer_references()` runs the original model once and caches, per pruned layer, its real-token inputs and outputs (half precision for half-precision models).
- `train_layer_biases()` trains all biases of one layer together with Adam on the layer's own device, evaluates train and validation loss at intervals, and skips steps where no token reaches a replaced expert.
- For top-1 routing, `closed_form_biases()` computes the exact optimum of the same objective as a reference.

Notes

- The objective is local: when several layers are pruned, earlier changes alter later layers' inputs, which this objective does not see.
- The learning rate must match the scale of the bias values.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`class LayerReference`	Cached original inputs/outputs of one layer (real tokens of all examples).
`.sequence(i)`	Inputs and outputs of example i.
`save_layer_reference(ref, cache_dir, meta)`	Write a layer's reference cache and its metadata.
`load_layer_reference(cache_dir, layer_id)`	Read a layer's reference cache and its metadata.
`cache_is_valid(cache_dir, layer_ids, expected_meta)`	True if cached references exist for all layers with matching metadata.
`capture_layer_references(model, adapter, layers, batches, device)`	Run the original model and keep (input, output) of each layer for real tokens.
`masked_sequence_batch(ref, rows, device, dtype)`	Right-pad sequences with zeros. Returns x [B,S,D], y [B,S,D], mask [B,S].
`reconstruction_loss(y_ref, y_hat, mask)`	(1/N) * sum over real tokens of ||y_ref - y_hat||_2^2, and N.
`evaluate_reconstruction(model, adapter, layer, ref, rows, device, batch_size)`	Layer MSE of the current biases on a set of examples (no gradients).
`closed_form_biases(model, adapter, layer, ref, rows, expert_ids, device, batch_size)`	Minimiser of the layer MSE on `rows`: b*_e = sum g^2 E / sum g^2.
`class LayerTrainingResult`	Losses, closed-form reference, bias changes and history of one layer.
`.to_dict()`	Convert to a plain dictionary for JSON reports.
`train_layer_biases(model, adapter, layer, expert_ids, ref, train_rows, val_rows, device, learning_rate, steps, batch_size, eval_interval, seed)`	Optimise all BiasExpert biases of one layer jointly with Adam.

Connections

Uses: `moe_pruning.model.architecture`, `moe_pruning.model.placement`, `moe_pruning.pruning.bias_expert`, `moe_pruning.utils.paths`

Used by: `scripts/train_biases.py`